CSE422; Assignment 02

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Date:

Answer 1

(a)

Have to prove, $P(Cheat \wedge Pass \mid Study) = P(Cheat \mid Study) * P(Pass \mid Study)$

$$LHS = P(Cheat \wedge Pass \mid Study)$$

$$= \frac{P(Cheat \wedge Pass \wedge Study)}{P(Study)}$$

$$= \frac{.25}{.45} = \frac{5}{9} = .556$$

$P(Study) = .25 + .15 + .02 + .03$
$= .45$

$P(Pass) = .25 + .15 + .10 + .13$
$= .63$

$P(Cheat) = .25 + .02 + .10 + .22$
$= .59$

$$RHS = P(Cheat \mid Study) * P(Pass \mid Study)$$

$$= \frac{P(Cheat \wedge Study)}{P(Study)} * \frac{P(Pass \wedge Study)}{P(Study)}$$

$$= \frac{.25 + .02}{.45} * \frac{.25 + .15}{.45}$$

$$= \frac{8}{15} = .533$$

Since, $LHS \neq RHS$

$\therefore$ Cheat and Pass are not conditionally independent given study.

b

$P(Pass \text{ or } Cheat) = P(Pass \vee Cheat) = P(Pass) + P(Cheat) - P(Pass \wedge Cheat)$

$= .63 + .59 - (.25 + .10)$

$= .87$

Answer 2

(a)

$P(\neg Cold) = .12 + .04 + .10 + .07$

$= .33$

(b)

$P(\neg Cloudy \mid \neg Rain \wedge \neg Cold)$

$= \dfrac{P(\neg Cloudy \wedge \neg Rain \wedge \neg Cold)}{P(\neg Rain \wedge \neg Cold)}$

$= \dfrac{.07}{.04 + .07}$

$= \dfrac{7}{11} = .636$

(c)

$P(\neg Rain \mid \neg Cloudy) = \dfrac{P(\neg Rain \wedge \neg Cloudy)}{P(\neg Cloudy)}$

$= \dfrac{.03 + .07}{.46}$

$= \dfrac{5}{23} = 0.217$

$P(\neg Cloudy) = .26 + .03 + .10 + .07$

$= .46$

$P(\neg Rain) = .06 + .04 + .03 + .07$

$= 0.2$

$P(Cloudy) = .32 + .06 + .12 + .04$

$= .54$

(d)

$P(\neg Rain \text{ or } Cloudy) = P(\neg Rain \vee Cloudy)$

$= P(\neg Rain) + P(Cloudy) - P(\neg Rain \wedge Cloudy)$

$= 0.2 + 0.54 - (.06 + .04)$

$= 0.64$

Answer 3

(a)

$$P(\text{Football} \mid \text{Left Hand}) = \frac{P(\text{Football} \wedge \text{Left Hand})}{P(\text{Left Hand})}$$

$$= \frac{\cdot 15}{\cdot 54}$$

$$= \frac{5}{18} = \cdot 278$$

$P(\text{Left Hand}) = \cdot 24 + \cdot 15 + \cdot 15$
$= \cdot 54$

$P(\text{Right Hand}) = \cdot 1 + \cdot 1 + \cdot 26$
$= \cdot 46$

(b)

$$P(\text{Right Hand} \mid \text{Cricket}) = \frac{P(\text{Right Hand} \wedge \text{Cricket})}{P(\text{Cricket})}$$

$$= \frac{\cdot 1}{\cdot 34} = \frac{5}{17} = \cdot 294$$

$P(\text{Cricket}) = \cdot 24 + \cdot 1$
$= \cdot 34$

(c)

$P(\text{Football} \wedge \text{Cricket}) = 0$

$P(\text{Football}) = \cdot 15 + \cdot 1$
$= \cdot 25$

(d)

$P(\text{Right Hand} \vee \text{Left Hand}) = \cancel{\cdot 24 +}\ \cdot 1 + \cdot 1 + \cdot 26 + \cdot 24 + \cdot 15 + \cdot 15$
$= 1$

(e) Have to check, $P(\text{Football} \wedge \text{Right Hand}) = P(\text{Football}) * P(\text{Right Hand})$

LHS $= P(\text{Football} \wedge \text{Right Hand})$
$= \cdot 1$

RHS $= P(\text{Football}) * P(\text{Right Hand})$
$= \cdot 25 * \cdot 46$
$= \cdot 115$

As, LHS $\neq$ RHS

Hence, Playing football does not depends on being Right-Handed.

Date:

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Answer 4

Combinations			Probability
X	Y	Z	
a	b	c	1/8
a	b	!c	3/8
a	!b	c	1/8
a	!b	!c	1/8
!a	b	c	0/8
!a	b	!c	0/8
!a	!b	c	2/8
!a	!b	!c	0/8

P(a∧b∧c)	c		!c	
	b	!b	b	!b
a	1/8	1/8	3/8	1/8
!a	0/8	2/8	0/8	0/8

$$P(!a \mid !b) = \frac{P(!a \wedge !b)}{P(!b)} = \frac{\frac{2}{8}+\frac{0}{8}}{\frac{4}{8}} = \frac{1}{2} = .5$$

(Answer)

Answer 5

Given, $P(\text{DetectLie}|\text{Lie}) = .96$

$P(\text{DetectTrue}|\text{True}) = .95$

$P(\text{Lie}) = .02$

Derived, $P(\text{True}) = 1 - .02 = .98$

$P(\text{DetectTrue}|\text{Lie}) = 1 - .96 = .04$

$P(\text{DetectLie}|\text{True}) = 1 - .95 = .05$

(a) $P(\text{Lie}|\text{DetectLie}) = ?$

$P(\text{True}|\text{DetectLie}) = ?$

$$P(\text{Lie}|\text{DetectLie}) = \frac{P(\text{DetectLie}|\text{Lie})\ P(\text{Lie})}{P(\text{DetectLie})} \quad \text{— (i)}$$

$$P(\text{True}|\text{DetectLie}) = \frac{P(\text{DetectLie}|\text{True})\ P(\text{True})}{P(\text{DetectLie})} \quad \text{— (ii)}$$

(i) ÷ (ii) =>

$$\frac{P(\text{Lie}|\text{DetectLie})}{P(\text{True}|\text{DetectLie})} = \frac{P(\text{DetectLie}|\text{Lie})\ P(\text{Lie})}{P(\text{DetectLie}|\text{True})\ P(\text{True})} = \frac{.96 \times .02}{.05 \times .98} = \frac{96}{245}$$

$$\Rightarrow \frac{P(\text{True}|\text{DetectLie})}{P(\text{Lie}|\text{DetectLie})} = \frac{245}{96} = 2.552$$

$\therefore P(\text{True}|\text{DetectLie}) = 2.552 \times P(\text{Lie}|\text{DetectLie})$

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We can see, $P(\text{True}|\text{DetectLie}) > P(\text{Lie}|\text{DetectLie})$

Hence, The person is likely not a liar.

(b)

$$P(\text{True}|\text{DetectLie}) = \frac{P(\text{DetectLie}|\text{True})\ P(\text{True})}{P(\text{Detect Lie})} \quad \text{—— ①}$$

$$P(\text{DetectLie}) = P(\text{Detect Lie} \wedge \text{True}) + P(\text{DetectLie} \wedge \text{Lie})$$

$$= P(\text{DetectLie}|\text{True})\ P(\text{True}) + P(\text{Detect Lie}|\text{Lie})\ P(\text{Lie})$$

$$= .05 * .98 + .96 * .02$$

$$= 0.0682$$

① =>

$$P(\text{True}|\text{DetectLie}) = \frac{0.05 \times .98}{.0682} = \frac{245}{341} = .7185 = 71.85\%$$

(Answer)

Answer 6

(a)

~~$P(P) = \frac{5}{8}$, $P(\neg P$~~

$P(\text{Tennis}) = \frac{5}{8}$, $P(\neg\text{Tennis}) = \frac{3}{8}$

Applying Naive Bayes,

$P(\text{Tennis} \mid \text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy}) = ?$

$$= \frac{P(\text{Tennis}|\text{Sunny}) * P(\text{Tennis}|\text{Mild}) * P(\text{Tennis}|\text{Normal}) * P(\text{Tennis}|\text{Windy}) * P(\text{Tennis})}{P(\text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy})} \quad \text{(I)}$$

$P(\neg\text{Tennis} \mid \text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy})$

$$= \frac{P(\neg\text{Tennis}|\text{Sunny}) * P(\neg\text{Tennis}|\text{Mild}) * P(\neg\text{Tennis}|\text{Normal}) * P(\neg\text{Tennis}|\text{Windy}) * P(\neg\text{Tennis})}{P(\text{Sunny} \wedge \text{Mind} \wedge \text{Normal} \wedge \text{Windy})} \quad \text{(II)}$$

Learning Phase

Outlook	Yes	No
Overcast	2/5	0/3
Sunny	2/5	3/3
Rainy	1/5	0/3

Humidity	Yes	No
Hot	0/5	1/3
Mild	3/5	2/3
Cool	2/5	0/3

Temp	Yes	No
Normal	4/5	0/3
High	1/5	3/3

Wind	Yes	No
True	3/5	1/3
False	2/5	2/3

Date:
(s) (s) (m) (t) (w) (t) (f)

(i) =>

$$P(\text{Tennis} \mid \text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy}) = \frac{\frac{2}{5} * \frac{3}{5} * \frac{4}{5} * \frac{3}{5} * \frac{5}{8}}{P(\text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy})}$$

$$= \frac{0.072}{P(\text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy})}$$

(ii) =>

$$P(\neg \text{Tennis} \mid \text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy}) = \frac{\frac{3}{3} * \frac{2}{3} * \frac{0}{3} * \frac{1}{3} * \frac{3}{8}}{P(\text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy})}$$

$$= 0$$

So, $P(\text{Tennis} \mid \text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy}) > P(\neg \text{Tennis} \mid \text{Sunny} \wedge \text{Mild} \wedge \text{Normal} \wedge \text{Windy})$

Hence, Yes the player is going to play tennis.

(b)

$$P(\text{Tennis} \mid \text{Overcast} \wedge \text{Hot}) = \frac{P(\text{Tennis} \mid \text{Overcast}) * P(\text{Tennis} \mid \text{Hot}) * P(\text{Tennis})}{P(\text{Overcast} \wedge \text{Hot})}$$

$$= \frac{\frac{2}{5} * \frac{0}{5} * \frac{5}{8}}{P(\text{Overcast} \wedge \text{Hot})}$$

Date

(b)

Let, $X =$ Overcast $\wedge$ Hot $\wedge$ Normal $\wedge$ Windy

Applying Naive Bayes,

$$P(Tennis|X) = \frac{P(Tennis|Overcast) * P(Tennis|Hot) * P(Tennis|Normal) * P(Tennis|Windy) * P(Tennis)}{P(X)}$$

$$= \frac{\frac{2}{5} * \frac{0}{5} * \frac{4}{5} * \frac{3}{5} * \frac{5}{8}}{P(X)}$$

$$= 0$$

$$P(\neg Tennis|X) = \frac{P(\neg Tennis|Overcast) * P(\neg Tennis|Hot) * P(\neg Tennis|Normal) * P(\neg Tennis|Windy) * P(\neg Tennis)}{P(X)}$$

$$= \frac{\frac{0}{3} * \frac{1}{3} * \frac{0}{3} * \frac{1}{3} * \frac{3}{8}}{P(X)}$$

$$= 0$$

As both are 0.

Hence, cannot determine.

Date:

Answer 7

x	Y	Y'
2	14	7
4	29	13
6	42	19

$Y' = mx + b$

$m = 3$

$b = 1$

$L.R = 0.01$

Mean Square Error,

$$\text{Loss function, } L = MSE = \frac{\sum_{i=1}^{n} (Y_i - Y_i')^2}{n} = \frac{(Y_1 - Y_1')^2 + (Y_2 - Y_2')^2 + (Y_3 - Y_3')^2}{3}$$

$$= \frac{(14-7)^2 + (29-13)^2 + (42-19)^2}{3}$$

$$= 278$$

~~Update m,~~

$$\frac{\partial L}{\partial m_1} = \frac{1}{3} \frac{\partial}{\partial m_1}$$

$$L = \frac{1}{3}\left\{(Y_1 - Y_1')^2 + (Y_2 - Y_2')^2 + (Y_3 - Y_3')^2\right\}$$

$$= \frac{1}{3}\left[\{Y_1 - (m_1 x_1 + b_1)\}^2 + \{Y_2 - (m_1 x_2 + b_1)\}^2 + \{Y_3 - (m_1 x_3 + b_1)\}^2\right]$$

Update m,

$$\frac{\partial L}{\partial m_1} = \frac{1}{3}\Big[2\{Y_1 - (m_1 x_1 + b_1)\}(-x_1) + 2\{Y_2 - (m_1 x_2 + b_1)\}(-x_2) +$$

$$2\{Y_3 - (m_1 x_3 + b_1)\}\cancel{x_3}(-x_3)\Big]$$

$$= \frac{2}{3}\left[\{Y_1 - (m_1 x_1 + b_1)\}(-x_1) + \{Y_2 - (m_1 x_2 + b_1)\}(-x_2) + \{Y_3 - (m_1 x_3 + b_1)\}\cancel{x_3}(-x_3)\right]$$

$$\cancel{= \frac{2}{3}[\{14 - 3 \times 2 +}$$

$$= \frac{2}{3}\left\{(Y_1 - Y_1')(-x_1) + (Y_2 - Y_2')(-x_2) + (Y_3 - Y_3')(-x_3)\right\}$$

Date
(s) (s) (m) (t) (w) (t) (f)

$= \frac{2}{3}\{(14-7)\times(-2) + (29-13)\times(-4) + (42-19)\times(-6)\}$

$= -144$

new m, $m_2 = m_1 -$ stepsize

$= m_1 - (\frac{\partial L}{\partial m_1} \times L.R)$

$= 3 - (-144 \times 0.01)$

$\therefore m_2 = \cancel{1.56}\ 4.44$

Update b,

$\cancel{\frac{\partial L}{\partial b_1} = \frac{2}{3}\{y_1 - (m_1x_1 + b_1x_1)\} + \{y_2 - (m_1x_2 + b_1)\} + \{y_3 - (m_1x_3 + b_1)\}}$

$\frac{\partial L}{\partial b_1} = \frac{2}{3}[\{y_1 - (m_1x_1 + b_1)\}(-1) + \{y_2 - (m_1x_2 + b_1)\}(-1) + \{y_3 - (m_1x_3 + b_1)\}(-1)]$

$= \frac{2}{3}\{(y_1 - y_1')(-1) + (y_2 - y_2')(-1) + (y_3 - y_3')(-1)\}$

$= \frac{2}{3}\{(14-7)(-1) + (29-13)(-1) + (42-19)(-1)\}$

$= -\frac{92}{3}$

$= -30.67$

new b, $b_2 = b_1 -$ stepsize

$= b_1 - (\frac{\partial L}{\partial b_1} \times L.R.)$

$= 1 - (-30.67 \times 0.01)$

$= 1.3067$

$\therefore m_2 = 4.44$ and $b_2 = 1.3067$

Date:

Answer 10

x	Y	Y'
1	5	3
3	12	7
5	15	11

$Y' = mx + b$

$m = 2$

$b = 1$

(a) Mean Square Error (Loss function, L),

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - Y_i') = \frac{(Y_1-Y_1')^2 + (Y_2-Y_2')^2 + (Y_3-Y_3')^2}{3}$$

$$= \frac{(5-3)^2 + (12-7)^2 + (15-11)^2}{3}$$

$$= 15$$

(b) $L.R = 0.05$

$$L = \frac{1}{3}\{(Y_1 - Y_1')^2 + (Y_2 - Y_2')^2 + (Y_3 - Y_3')^2\}$$

$$= \frac{1}{3}\left[\{Y_1 - (m_1x_1+b_1)\}^2 + \{Y_2 - (m_1x_2+b_1)\}^2 + \{Y_3 - (m_1x_3+b_1)\}^2\right]$$

1st iteration,

$$\frac{\partial L}{\partial m_1} = \frac{2}{3}\left[\{Y_1 - (m_1x_1+b_1)\}(-x_1) + \{Y_2 - (m_1x_2+b_1)\}(-x_2) + \{Y_3 - (m_1x_3+b_1)\}(-x_3)\right]$$

$$= \frac{2}{3}\{(Y_1 - Y_1')(-x_1) + (Y_2 - Y_2')(-x_2) + (Y_3 - Y_3')(-x_3)\}$$

$$= \frac{2}{3}\{(5-3)(-1) + (12-7)(-3) + (15-11)(-5)\}$$

$$= -\frac{74}{3}$$

$$= -24.67$$

$$\frac{\partial L}{\partial b_1} = \frac{2}{3}\left[\{y_1-(m_1x_1+b_1)\}(-1) + \{y_2-(m_1x_2+b_1)\}(-1) + \{y_3-(m_1x_3+b_1)\}(-1)\right]$$

$$= \frac{2}{3}\{(y_1-y_1')(-1) + (y_2-y_2')(-1) + (y_3-y_3')(-1)\}$$

$$= \frac{2}{3}\{(5-3)(-1) + (12-7)(-1) + (15-11)(-1)\}$$

$$= -\frac{22}{3}$$

$$= -7.33$$

---

new m, $m_2 = m_1 - \left(\frac{\partial L}{\partial m_1} \times LR\right) = 2 - (-24.67 \times 0.05)$

$= 3.2335$

new b, $b_2 = b_1 - \left(\frac{\partial L}{\partial b_1} \times LR\right) = 1 - (-7.33 \times 0.05)$

$= 1.3665$

2nd iteration,

x	Y	Y'
1	5	4.6
3	12	11.067
5	15	17.534

$m = 3.2335$

$b = 1.3665$

$Y' = mx + b$

$$\frac{\partial L}{\partial m_2} = \frac{2}{3}\{(y_1-y_2')(-x_1) + (y_2-y_2')(-x_2) + (y_3-y_3')(-x_3)\}$$

$$= \frac{2}{3}\{(5-4.6)(-1) + (12-11.067)(-3) + (15-17.534)(-5)\}$$

$$= 6.314$$

Date:

$$\frac{\partial L}{\partial b_2} = \frac{2}{3}\{(y_1 - y_1')(-1) + (y_2 - y_2')(-1) + (y_3 - y_3')(-1)\}$$

$$= \frac{2}{3}\{(5-4.6)(-1) + (12-11.067)(-1) + (15-17.534)(-1)\}$$

$$= \frac{1201}{1500}$$

$$= 0.80067$$

new m, $m_3 = m_2 - (\frac{\partial L}{\partial m_2} \times LR) = 3.2335 - (6.374 \times 0.05)$

$$= 2.9178$$

new b, $b_3 = b_2 - (\frac{\partial L}{\partial b_2} \times LR) = 1.3665 - (0.80067 \times 0.05)$

$$= 1.3265$$

(C)

For i. $L_1 = (y_1 - y_1')^3 + (y_2 - y_2')^3 + (y_3 - y_3')^3$

$$= (5-3)^3 + (12-7)^3 + (15-11)^3$$

$$= 197$$

For ii. $L_2 = |y_1 - y_1'| + |y_2 - y_2'| + |y_3 - y_3'|$

$$= |5-3| + |12-7| + |15-11|$$

$$= 11$$

"ii" is better be cause it ~~gives~~ always gives positive value, which is easy to interphete. ~~where as~~ "i" can some time generate negative value. If that happens then there is a chance of positive and negative value given by "i" can some time cancel each other and show 0 error even after having error.

Therefor "ii" is better.

Date

Answer 11

Given, $x_1 = 1$, $x_2 = 2$, $Y = 0$, $w_1 = 0.5$, $w_2 = -0.25$, $b = 0$, $\alpha = 0.1$

A

$$z = w_1 x_1 + w_2 x_2 + b_1 = 0.5 \times 1 + (-0.25) \times 2 + 0$$

$$= 0$$

$$Y' = \sigma(0) = \frac{1}{1+e^{-0}} = 0.5$$

B

$$\frac{\partial Loss}{\partial w_1} = (Y' - Y)\, x_1 = (0.5 - 0) \times 1 = 0.5$$

new w_1, $$w_{1_{new}} = w_1 - \left(\frac{\partial Loss}{\partial w_1} \times \alpha\right)$$

$$= 0.5 - (0.1 \times 0.5)$$

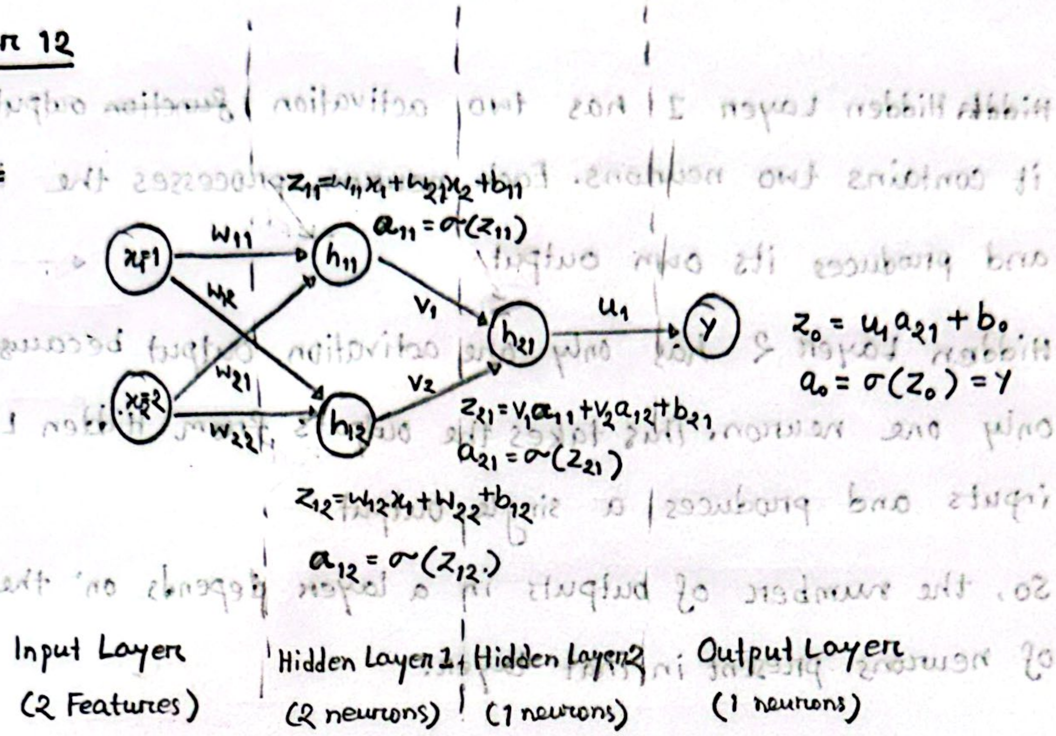
$$= 0.45$$

C

Decaying learning rate means the learning rate decreases gradually during training. In the beginning a larger learning rate helps the model learn quickly and make faster progress. As training continuous the learning rate becomes smaller. Which helps the model to make more precise and stable updates. This prevent the model from overshooting the minimum loss and improves the overall convergence of logistic regression.

Answer 12

A.



B

Given, $a_{11} = 0.5$ $\quad a_{12} = 0.5$ $\quad V_1 = 0.1$ $\quad V_2 = 0.5$

$$z_{21} = v_1 a_{11} + v_2 a_{12} + b_{21}$$

$$= 0.5 \times 0.1 + 0.5 \times 0.5 + 0$$ [let $b_{21} = 0$, As not given]

$$= 0.30$$

$$a_{21} = \sigma(z_{21}) = \sigma(0.30) = \frac{1}{1+e^{-0.30}}$$

$$= 0.5744$$

(Answer)

Date:

s s m t w t f

c

~~Hidda~~ Hidden Layer 1 has two activation ~~function~~ output because it contains two neurons. Each neuron processes the input separately and produces its own output.

Hidden Layer 2 has only one activation output because it contains only one neuron. This takes the outputs from Hidden Layer 1 as inputs and produces a single output.

So, the number of outputs in a layer depends on the number of neurons present in that layer.